

Experiment-18

Aim

To perform edge detection along the Y-axis of an image using the Sobel operator and identify horizontal edges present in the image.

Procedure

1. Select and load a grayscale image.
2. Read the image using OpenCV.
3. Convert the image to grayscale if required.
4. Define the Sobel Y-axis kernel.
5. Apply the kernel to the image using convolution.
6. Calculate the absolute magnitude of the resulting image.
7. Normalize the pixel values to the range 0–255.
8. Display the original and Y-axis edge-detected images.
9. Observe the horizontal edges highlighted in the output.

Python Code

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
img = cv2.imread("image.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)
sobel_y = cv2.convertScaleAbs(sobel_y)
plt.figure(figsize=(10, 5))
plt.subplot(1, 2, 1)
plt.imshow(gray, cmap='gray')
plt.title("Original Image")
plt.axis("off")

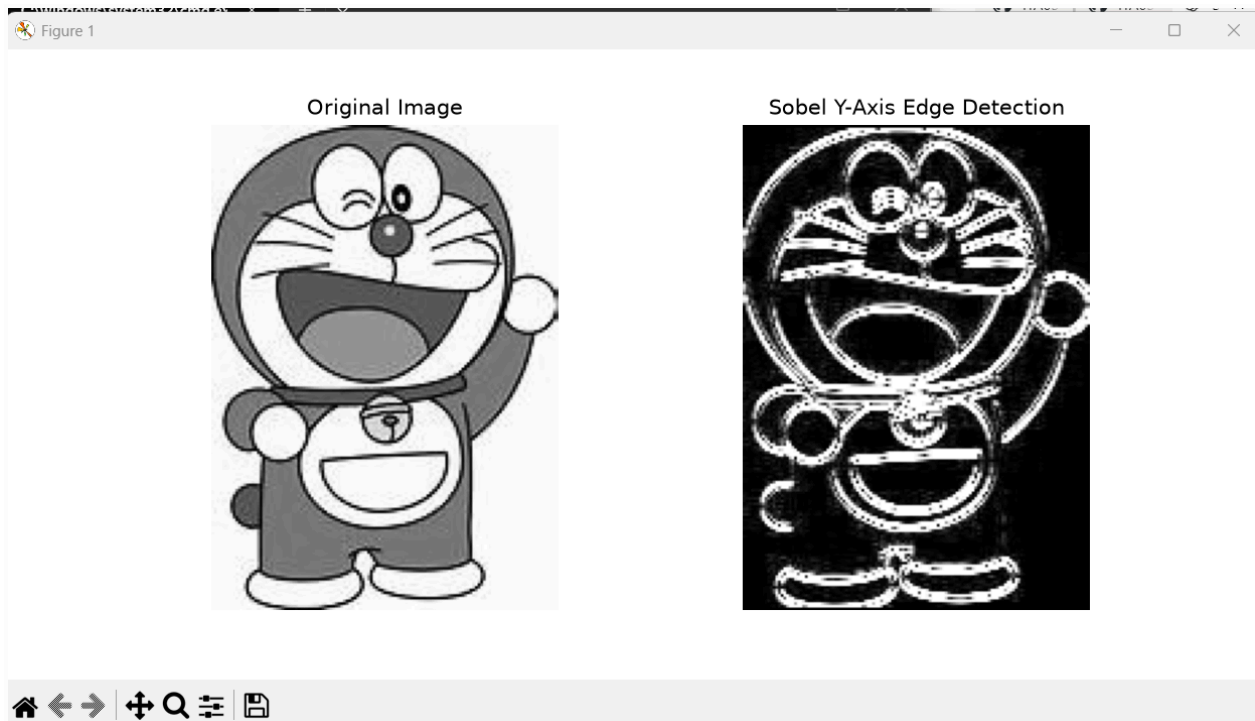
plt.subplot(1, 2, 2)
```

```
plt.imshow(sobel_y, cmap='gray')
plt.title("Sobel Y-Axis Edge Detection")
plt.axis("off")
plt.show()
```

Sample Input:



Output:



Results:

The Sobel operator along the Y-axis was successfully applied to the input image. The resulting image highlights the horizontal edges, where there is a significant change in intensity in the vertical direction.